ECN 594: Product Differentiation and Hotelling

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Recap: Cournot and Bertrand

- Last time we covered:
  - **Cournot:** Firms choose quantities; $L = s_i / |\varepsilon|$
  - **Bertrand:** Firms choose prices; $P = MC$ (paradox!)
- **Escaping the Bertrand paradox:**
  1. Capacity constraints (Kreps-Scheinkman)
  2. **Product differentiation** (today!)
  3. Repeated interaction (collusion, later)
- Today: How differentiation creates pricing power

Plan

1. **Product differentiation: why it matters**
2. Hotelling model
3. Connection to demand estimation

Why differentiation matters

- Bertrand paradox: $P = MC$ with homogeneous products
- **Solution:** Product differentiation!
- If products are different, consumers don't all buy from lowest-price firm
- Firms have some pricing power
- This is exactly what we modeled in Part 1 (logit demand)
- Now: a classic spatial model of differentiation

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1. Product differentiation: why it matters
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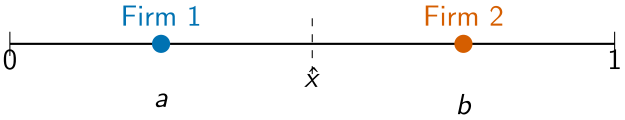
Hotelling model: setup

- Consumers uniformly distributed on $[0, 1]$ ("Main Street")
- Two firms located at positions a and b on $[0, 1]$
- Consumer at location x has utility:

$$u_j = v - p_j - t|x - \ell_j|$$

- v: base value of product
- p_j: price of firm j
- t: transport cost per unit distance
- $|x - \ell_j|$: distance to firm j

Hotelling: graphical intuition



- Consumers to the left of $\hat{x}$ buy from Firm 1
- Consumers to the right of $\hat{x}$ buy from Firm 2
- $\hat{x}$ is the "indifferent consumer"

Finding the indifferent consumer

- Consumer at $\hat{x}$ is indifferent between firms:

$$v - p_1 - t|\hat{x} - a| = v - p_2 - t|b - \hat{x}|$$

- With $a = 0$ and $b = 1$ (firms at endpoints):

$$\begin{aligned} v - p_1 - t\hat{x} &= v - p_2 - t(1 - \hat{x}) \\ p_2 - p_1 &= t(1 - 2\hat{x}) \\ \hat{x} &= \frac{1}{2} + \frac{p_2 - p_1}{2t} \end{aligned}$$

- Demand for firm 1: $D_1 = \hat{x}$

- Demand for firm 2: $D_2 = 1 - \hat{x}$

Hotelling: equilibrium prices

- Firm 1 maximizes: $\pi_1 = (p_1 - c) \cdot \hat{x}(p_1, p_2)$
- FOC: $\frac{\partial \pi_1}{\partial p_1} = \hat{x} + (p_1 - c)\frac{\partial \hat{x}}{\partial p_1} = 0$
- With $\frac{\partial \hat{x}}{\partial p_1} = -\frac{1}{2t}$:

$$\frac{1}{2} + \frac{p_2 - p_1}{2t} - \frac{p_1 - c}{2t} = 0$$

- Symmetric equilibrium $(p_1 = p_2 = p^*)$:

$$p^* = c + t$$

- **Markup = transport cost!**

Hotelling: interpretation

- $p^* = c + t$: Firms charge above marginal cost
- **Transport cost t measures differentiation**
  - High t: products very different $\rightarrow$ high markup
  - Low t: products similar $\rightarrow$ low markup
  - $t \rightarrow 0$: products identical $\rightarrow$ Bertrand ($p \rightarrow c$)
- **No Bertrand paradox:** Differentiation creates pricing power
- Each firm gets half the market: $D_1 = D_2 = 1/2$
- Profit: $\pi = (p^* - c) \cdot \frac{1}{2} = \frac{t}{2}$

Worked example: Hotelling

- **Question:** Two ice cream vendors on a beach of length 1 mile. Transport cost $t = 2$ dollars per mile. Marginal cost $c = 1$.
- (a) Find the equilibrium price.
- (b) If firm 1 raises price to $p_1 = 4$, what is its market share?
- (c) Calculate firm 1's demand elasticity at the equilibrium.

Take 4 minutes.

Worked example: Hotelling (solution)

- **(a)** $p^* = c + t = 1 + 2 = 3$

- **(b)** At $p_1 = 4$, $p_2 = 3$:

$$\hat{x} = \frac{1}{2} + \frac{3-4}{2(2)} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

- Firm 1's market share falls to 25%

- **(c)** At equilibrium: $D_1 = 1/2$, $\frac{\partial D_1}{\partial p_1} = -\frac{1}{2t} = -\frac{1}{4}$

$$\varepsilon_1 = \frac{\partial D_1}{\partial p_1} \cdot \frac{p_1}{D_1} = -\frac{1}{4} \cdot \frac{3}{1/2} = -1.5$$

Hotelling: why transport cost matters

- **High t:** Strong differentiation
  - Consumers have strong location preferences
  - Firm can raise price without losing many customers
  - Example: specialty restaurants vs fast food
- **Low t:** Weak differentiation
  - Consumers nearly indifferent across locations
  - Small price cut steals many customers
  - Approaches Bertrand as $t \to 0$
- In demand estimation: $1/|\alpha|$ plays similar role to t

Practice: T/F on Hotelling

- **True, False, or NEI:**
- (a) In Hotelling equilibrium, firms always split the market equally.
- (b) A monopolist in Hotelling should locate at the center of the line.
- (c) Higher transport costs benefit consumers because products are more differentiated.

Take 2 minutes.

Practice: T/F on Hotelling (solution)

- **(a) TRUE** (with caveat). If firms are at endpoints and have same costs, yes. With asymmetric locations or costs, shares differ.
- **(b) TRUE.** Monopolist minimizes total transport costs by locating at center. This maximizes consumer value and thus WTP.
- **(c) FALSE.** Higher t means higher prices ($p^* = c + t$) and higher transport costs. Both hurt consumers.

Welfare in Hotelling

- **Total welfare** = Consumer surplus + Profits
- Transport costs are deadweight loss
- **Socially optimal locations:** $a = 1/4$, $b = 3/4$
  - Minimizes total transport costs
- **Equilibrium locations:** Both firms at $1/2$ (minimum differentiation)
  - Firms want to capture more customers
  - But this increases total transport costs
- "Principle of minimum differentiation" (but fragile)

Plan

1. Product differentiation: why it matters
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3. **Connection to demand estimation**

Connection to demand estimation

- Hotelling is a specific **differentiated Bertrand** model
- Location $\leftrightarrow$ product characteristics
- Transport cost $\leftrightarrow$ preference heterogeneity
- **Logit demand** generalizes this idea:
  - Products differ in characteristics space
  - Consumers have heterogeneous preferences
  - Price competition with differentiated products
- Next lectures: how to use demand estimates for merger simulation

From Hotelling to logit: the connection

Hotelling	**Logit**
Location on line	Product characteristics
Transport cost t	Sensitivity $1/|\alpha|$
Consumer position	Consumer preferences
Linear utility	Logit choice probabilities
2 products	J products

- Logit demand lets us do empirical Hotelling-style analysis
- Estimate α from data $\rightarrow$ calibrate "differentiation"

Oligopoly models: summary

- **Homogeneous products:**
  - Cournot (quantity): $L = s_i / |\varepsilon|$
  - Bertrand (price): $P = MC$
- **Differentiated products:**
  - Hotelling/Differentiated Bertrand: $P > MC$
  - Markup depends on differentiation (transport cost)
- **Empirical approach:**
  - Estimate demand (logit)
  - Assume pricing behavior (usually differentiated Bertrand)
  - Back out marginal costs

Preview: merger simulation

- **Key question:** If two firms merge, what happens to prices?
- **Approach:**
  1. Estimate demand $\rightarrow$ get elasticities
  2. Assume Bertrand pricing $\rightarrow$ back out MC
  3. Change ownership structure
  4. Solve new Bertrand equilibrium
  5. Compare prices and welfare
- This is exactly what HW2 will do!
- Will cover in detail in Lecture 10

Key Points

1. **Bertrand paradox:** $P = MC$ with homogeneous products
2. **Product differentiation** creates pricing power
3. **Hotelling:** $p^* = c + t$ (markup = transport cost)
4. Higher t (more differentiation) $\rightarrow$ higher markup
5. Indifferent consumer: $\hat{x} = \frac{1}{2} + \frac{p_2 - p_1}{2t}$
6. Welfare: socially optimal locations differ from equilibrium
7. Hotelling connects to logit demand from Part 1
8. Transport cost $t \leftrightarrow$ price sensitivity $1/|\alpha|$